

 Part 1 – Question Bank

10-Mark Questions

1. Explain the various phases of the Data Mining Process in detail.
2. Describe the four major Data Mining Techniques and their applications.
3. Detail the different stages and methods involved in Data Preprocessing.
4. Explain the theoretical framework, principles, and procedures of the Apriori Algorithm.
5. Discuss the Frequent Pattern (FP) Growth Algorithm, its tree structure, and its advantages over Apriori.

5-Mark Questions

6. Define Data Mining and list its key characteristics and takeaways.
7. Provide a detailed comparison between Data Mining and Machine Learning.
8. Describe the various applications of Data Mining across different industries.
9. Explain the different strategies for handling missing values during data cleaning.
10. Elaborate on binning as a method for smoothing noisy data.
11. Compare normalization and standardization as data transformation methods.
12. Discuss the three main categories of Data Reduction techniques.
13. Explain Link Analysis and its three specific specializations.
14. Define the metrics used to evaluate the strength of association rules.
15. Outline the pros and cons of using Data Mining in an organization.

2-Mark Questions

16. What is the Knowledge Discovery in Databases (KDD) process?
17. Define Predictive Modeling.
18. What is the entity identification problem in data integration?
19. Define Database Segmentation.
20. State the Apriori Principle.
21. What constitutes a strong rule in association mining?
22. Define a Frequent Itemset.
23. What is Deviation Detection?

EXAM POV NOTES (Q&A FORMAT)**✦ Part 2 – Complete Answer Bank****10-Mark Questions**

1. Explain the various phases and steps involved in the Data Mining Process (KDD).

Ans: The Data Mining Process, or Knowledge Discovery in Databases (KDD), is an iterative sequence of stages used to extract valid, novel, and actionable knowledge from raw data sources.

- **Problem Definition Phase:** This initial step focuses on understanding project requirements and business objectives from a stakeholder perspective. It involves translating a vague business goal, such as increasing sales, into a defined data mining problem to guide the project.
- **Data Understanding Phase:** During this stage, data is collected from available sources to identify initial insights and potential quality issues. Technical activities like data loading and preliminary integration are performed to ensure the collection is proper for analysis.
- **Data Preparation Phase:** This is the most resource-intensive phase, typically consuming about 90% of the total project time. It involves selecting necessary data, cleaning it of errors, and formatting it into the specific structure required for mining algorithms.
- **Modeling Phase:** In this stage, various data mining algorithms are selected and applied to the prepared dataset to build representative models. Different techniques may be tested on the same data to see which best achieves the project's predictive or descriptive objectives.
- **Evaluation Phase:** After a model is built, its results are assessed to determine if they truly satisfy the originally stated business goals. To ensure reliability, the data is usually split into separate training and testing sets to validate accuracy.
- **Deployment Phase:** The final stage involves presenting derived insights in a format that stakeholders can easily use for decision-making. Depending on the project, this can range from simple report generation to setting up a complex, iterative system.
- **Data Cleaning:** As a sub-process of preparation, data cleaning involves filling in missing values and smoothing out noisy data to enhance quality. It resolves inconsistencies and identifies outliers to ensure the subsequent mining results are not skewed by erroneous data.
- **Data Integration:** This process combines data from multiple disparate sources, such as different databases or flat files, into a unified store. It addresses issues like entity identification, ensuring that different labels from various sources refer to the same physical object.
- **Pattern Evaluation:** This step involves identifying the truly interesting and useful patterns from the total set generated by the model. It filters out trivial or previously known information to focus solely on novel and valid knowledge for the user.
- **Knowledge Representation:** This is the final output step where findings are visualised using dashboards or Graphical User Interfaces. It transforms complex mathematical results into an understandable format that allows stakeholders to take action.

2. Describe the four major Data Mining Techniques and their practical applications.

Ans: Data mining techniques are the core methodologies used to discover different types of patterns and knowledge within large-scale organisational databases.

- **Predictive Modeling:** This technique uses known historical results to create and validate a model that forecasts future outcomes. It is a fundamental tool in predictive analytics that attempts to answer what might happen in the future based on past data.
- **Classification:** A subset of predictive modeling used to assign items in a collection to specific target categories or classes. Practical applications include predicting student placement, sports match outcomes, or the future sale price of a property.
- **Database Segmentation:** Based on the concept of clustering, this technique falls under unsupervised learning where data points are not pre-labelled. It identifies groups of similar records that share a number of common properties or attributes.
- **Customer Segmentation:** A primary application of segmentation used to group customers with similar buying habits for targeted marketing. It helps businesses understand their diverse customer base to improve service delivery and reduce customer churn.
- **Link Analysis:** This technique aims to establish associations between individual records or sets of records within a large database. It identifies relationships where the presence of one item implies the presence of another in the same event.
- **Associations Discovery:** A specialisation of link analysis that locates items frequently purchased together in a single transaction. It is famously used for Market Basket Analysis to help retailers identify cross-selling opportunities.
- **Sequential Pattern Discovery:** This branch of link analysis finds patterns between events occurring over a period of chronological time. It determines if the presence of one set of items is consistently followed by another set in a database of events.
- **Time Sequence Discovery:** This method is used to determine if links exist between two sets of data that are fundamentally time-dependent. A common practical application is in the finance industry for predicting stock price movements over time.
- **Deviation Detection:** This technique focuses on identifying outliers that represent a significant departure from known norms or expectations. It uses statistics and visualisation to highlight records that do not fit the general pattern of the data.
- **Fraud Detection:** An essential application of deviation detection used to identify suspicious activities in credit card usage or insurance claims. It allows organisations to prevent financial losses by flagging anomalies that suggest fraudulent behaviour.

3. Detail the different methods and stages involved in Data Preprocessing.

Ans: Data preprocessing is a critical prerequisite for data mining that involves transforming raw data into a clean, understandable, and error-free format.

- **Data Cleaning Stage:** This first stage involves resolving issues like missing values, noisy data, and inconsistencies within the raw dataset. It is essential because the quality of the input data directly affects the accuracy of the final mining results.
- **Handling Missing Values:** Missing data points can be managed by manually filling them, using global constants, or applying the attribute mean. In extreme cases where a record is too incomplete, the entire tuple may be ignored to maintain integrity.

- **Smoothing Noisy Data:** Noise represents random errors or unwanted variance in a measurable variable that can adversely affect algorithms. Techniques like binning, regression, and clustering are used to "smooth" the data and remove these fluctuations.
- **Data Integration Stage:** This process combines data from multiple disparate sources, such as databases and flat files, into a unified store. It ensures that all relevant data is consolidated for a comprehensive analysis of the business problem.
- **Entity Identification Resolution:** During integration, analysts must ensure that different identifiers, like "student_id" and "student_number," refer to the same entity. Metadata is used as a reference to avoid errors and ensure correct mapping during this consolidation.
- **Data Transformation Stage:** This involves consolidating or transforming data into standard forms that are better suited for specific mining algorithms. It often converts data from a source format into a destination format required by a data warehouse.
- **Normalization Method:** This transformation technique converts all attribute values into a specific fixed range, typically between 0 and 1. It is useful for algorithms requiring consistent boundaries but can be sensitive to extreme outliers.
- **Standardization Method:** In this method, values are spread out so that the resulting dataset has a standard deviation of 1. It is the preferred transformation strategy when the data contains significant outliers that might distort normalization.
- **Data Reduction Stage:** This process reduces the volume or dimensions of a dataset without compromising the integrity of the original information. It makes the mining process more efficient and often results in better accuracy for large datasets.
- **Dimension Reduction:** This strategy removes redundant or weakly relevant attributes from the dataset using specific algorithms like PCA. It ensures the model focuses only on the most important variables, reducing computational complexity.

4. Explain the theoretical framework, principles, and procedures of the Apriori Algorithm.

Ans: The Apriori Algorithm is a fundamental method for association rule mining that uses a two-step approach to find frequent itemsets in a database.

- **Frequent Itemset Generation:** The first and most computationally expensive step involves finding all itemsets that meet a minimum support threshold. This process requires a full database scan to count the frequency of occurrences for every item.
- **Rule Generation:** Once frequent itemsets are identified, the algorithm lists all possible association rules derived from them. Support and confidence are calculated for each rule, and those failing to meet thresholds are pruned.
- **The Apriori Principle:** The algorithm is built on the principle that if an itemset is frequent, then all of its subsets must also be frequent. This logic allows the algorithm to significantly reduce the number of itemsets it needs to test.
- **Infrequent Pruning:** Conversely, if an itemset is found to be infrequent, all of its supersets are guaranteed to be infrequent. These supersets are pruned and never generated, which optimizes the search through the itemset lattice.

- **General-to-Specific Strategy:** Apriori uses a breadth-first search strategy where pairs of frequent (k-1) itemsets are merged to obtain candidate k-itemsets. It progressively moves from single items to larger combinations as long as they remain frequent.
- **Candidate List Generation:** In each iteration, the algorithm starts with individual items called candidates and calculates their support counts. Only those that satisfy the "min_sup" threshold are carried over to the next level of testing.
- **Breadth-First Traversal:** The algorithm traverses the itemset lattice level by level, discovering all frequent 1-itemsets, then 2-itemsets, and so on. This systematic approach ensures that no potential frequent itemset is overlooked during the scan.
- **Market Basket Application:** The algorithm is primarily used to identify items that "go together" in shopper transactions. It generates rules like {Diapers} -> {Beer} to help retailers understand customer purchasing patterns.
- **Support Metric:** Support measures the fraction of transactions that contain a particular itemset. It is used to eliminate uninteresting rules that occur too rarely to be profitable for a business.
- **Confidence Metric:** Confidence measures the reliability of the inference made by a rule by showing how often item B is bought with A. High confidence indicates a strong predictive relationship between the items in the rule.

5. Discuss the Frequent Pattern (FP) Growth Algorithm, its structure, and mining steps.

Ans: The FP Growth Algorithm is an efficient improvement over Apriori that finds frequent patterns without the need for costly candidate generation.

- **Tree Representation:** The algorithm represents the entire database in a compact tree structure called a Frequent Pattern Tree (FP Tree). This structure maintains the association between itemsets and significantly reduces memory usage.
- **Fragmented Processing:** Unlike Apriori, the database is fragmented using one frequent item into "pattern fragments". These fragments are then analysed individually, allowing for a more focused and faster mining process.
- **Root and Node Structure:** The FP Tree starts with a null root node, while lower nodes represent specific items from the transactions. Each node maintains a count of how many times that item appears in that specific path.
- **Descending Order Sort:** Before constructing the tree, items in the database are sorted in descending order based on their frequency. This sorting ensures that the most common items are closer to the root, maximizing path sharing.
- **Shared Prefix Logic:** If multiple transactions share common items, their branches in the FP Tree share a common prefix to the root. This sharing effectively compresses the database information into a much smaller representation.
- **Initial Database Scan:** The first step involves scanning the database to find the total occurrences of each item, similar to the start of Apriori. This scan determines which items meet the minimum support threshold to be included.

- **Construction Process:** A second scan is performed to examine each transaction and add it to the FP Tree based on the sorted frequent items. As items are added to existing paths, their individual counts are incremented accordingly.
- **Mining the Tree:** Mining begins from the lowest nodes in the tree, which represent frequent patterns of length one. The algorithm then traverses the paths upward toward the root to find longer combinations.
- **Conditional FP Tree:** For each item, a conditional FP Tree is formed by considering only the itemsets that appear in its path. This sub-tree is used to generate the final frequent patterns associated with that specific item.
- **Speed Advantage:** Because FP Growth only requires two database scans and no candidate generation, it is significantly faster than Apriori. Its runtime increases linearly rather than exponentially as the number of itemsets grows.

5-Mark Questions

6. Define Data Mining and list its key takeaways from an organisational perspective.

Ans: Data mining is the process of extracting previously unknown, valid, and actionable patterns from massive datasets.

- **Automated Discovery:** It is an automated process designed to find hidden patterns in large volumes of historical data, often stored in a data warehouse.
- **Volume Handling:** The technique is specifically designed to handle massive data scales, ranging from Gigabytes to Zettabytes in big data scenarios.
- **Actionable Knowledge:** The discovered patterns must be valid and understandable so that they can be used directly in an enterprise's decision-making process.
- **Future Prediction:** A key organisational value is its ability to determine historical patterns to accurately predict future customer behaviour.
- **Accuracy Scale:** While mining is possible with small datasets, the accuracy of its predictions increases significantly as the volume of data grows.

To provide you with a complete and comprehensive question bank, here are the detailed theory-focused answers for all remaining questions (Q6 to Q23) from your list, based strictly on the provided sources.

7. Provide a detailed comparison between Data Mining and Machine Learning.

Data mining and machine learning are distinct yet related fields that focus on data extraction and algorithmic learning respectively.

Basic for Comparison	Data Mining	Machine Learning
Meaning	Involves extracting useful knowledge from a large amount of existing data.	Introduces new algorithms from data as well as past experience.
Responsibility	Used to examine patterns in existing data to set business rules.	Teaches the computer how to learn and understand the given rules.

Nature	Involves significant human involvement and manual intervention.	It is automated and self-implementing with very little human effort.
History	Initially called knowledge discovery in databases and started in 1930.	It was introduced later as a separate field of study in 1959.
Focus	Focuses on the discovery of previously unknown properties in data.	Focuses on predicting outcomes based on known properties learned.

8. Discuss the various Applications of data mining in modern industries.

Data mining transforms raw data into valuable information across various sectors to increase revenue and improve operational efficiency.

- **Finance and Banking:** Banks use mining for loan or credit card approvals by analyzing applicant risk and for detecting fraudulent financial transactions.
- **Retail and Marketing:** It is used for market segmentation to identify distinct customer groups and for direct marketing to improve brand loyalty.
- **E-Commerce Strategy:** Companies like Amazon use link analysis for product recommendations and market basket analysis to see which items go together.
- **Healthcare Industry:** In medical fields, it is applied to identify correlations between different diseases and to trace defects in treatment protocols.
- **Customer Retention:** Businesses use it for customer churn analysis to predict which users are likely to stop using a service and stay ahead of the competition.

9. Explain the different strategies for Handling Missing Values during data cleaning.

Real-world data is often incomplete, and data cleaning resolves this by filling in gaps to ensure the mining process is free from errors.

- **Manual Entry:** The data analyst can manually fill in missing values based on external knowledge, though this is difficult for massive datasets.
- **Global Constants:** A simple constant or a label like "Unknown" can be used to replace all missing values in a specific attribute.
- **Attribute Mean:** The algorithm calculates the mean (average) of all available values for that attribute and uses it to fill the missing spots.
- **Probability Estimation:** Analysts can use values that are statistically high in probability to guess and fill the missing data points.
- **Tuple Deletion:** If a record (tuple) has too many missing attributes, the entire row can be ignored to prevent biased results in the final model.

10. Elaborate on Binning as a method for smoothing noisy data.

Binning is a smoothing technique that sorts data and consults the "neighborhood" of values to reduce random error or unwanted variance.

- **Equal-Depth Partitioning:** The sorted data is distributed into a specific number of intervals called "bins" or "buckets," each containing approximately the same number of samples.

- **Smoothing by Bin Means:** In this method, every individual value within a bin is replaced by the calculated mean (average) of that specific bin.
- **Smoothing by Bin Medians:** Each value in a bin is replaced by the median value of that bin, which is often more robust to outliers.
- **Smoothing by Bin Boundaries:** The minimum and maximum values of a bin are identified as boundaries, and every other value in the bin is replaced by the closest boundary.
- **Noise Reduction:** By applying these neighborhood-based methods, the random fluctuations in data are removed, making the patterns clearer for mining algorithms.

11. Compare Normalization and Standardization as data transformation methods.

Normalization and standardization are techniques used to consolidate data into forms suited for mining, especially when dealing with different scales.

Feature	Normalization	Standardization
Output Range	Converts all attribute values to a fixed range, typically between 0 and 1.	Spreads out values so that the resulting dataset has a standard deviation of 1.
Outlier Handling	Extremely sensitive; an outlier can "crunch" all other values down toward zero.	Preferred method when data has outliers as it handles extreme values better.
Process	Scales the data based on the minimum and maximum values of the attribute.	Centers the data around the mean based on the spread of the values.
Goal	Used to ensure all features contribute equally to the distance calculations.	Used to make the data follow a standard normal distribution for certain algorithms.
Usage Rule	Use when you know the boundaries and have no significant outliers.	Use when the data is inconsistent or contains significant erroneous outliers.

12. Discuss the strategies used for Data Reduction in large datasets.

Data reduction creates a smaller representation of a dataset without compromising the integrity of the original data, leading to better accuracy and efficiency.

- **Dimension Reduction:** This strategy removes redundant or weakly relevant attributes from the dataset using algorithms like Principal Component Analysis (PCA).
- **Numerosity Reduction:** It is a technique used to choose smaller forms of data representation to physically reduce the volume of the dataset.
- **Data Compression:** Encoding mechanisms like Huffman coding are used to reduce dataset size through lossy or lossless compression techniques.
- **Efficiency Gains:** Carrying out mining on reduced datasets proves to be more efficient because the algorithm has to process fewer data points.
- **Integrity Maintenance:** The core goal is to ensure that the reduced data still yields the same useful knowledge as the original massive data store.

13. Explain Link Analysis and its three specific specialisations.

Link analysis aims to establish associations between individual records or sets of records within a database to find what "goes together".

- **Associations Discovery:** This specialization identifies items that imply the presence of other items during the same single event, such as products in a market basket.
- **Sequential Pattern Discovery:** It finds patterns between events occurring over time, where the presence of one set of items is followed by another set later on.
- **Time Sequence Discovery:** This method determines if links exist between two sets of data that are fundamentally time-dependent, such as stock price movements.
- **Business Applications:** Organizations use link analysis for building recommendation systems, market basket analysis, and direct marketing strategies.
- **Technical Goal:** The primary technical objective is to locate hidden relationships between disparate data points to predict future occurrences.

14. Define the Metrics used to evaluate the strength of association rules.

Association rule mining uses metrics like support and confidence to judge whether a discovered rule is interesting and reliable for business.

- **Support Metric:** Support is the fraction of transactions that contain a particular itemset; it measures how often a rule is applicable to the data.
- **Pruning Uninteresting Rules:** Support is used to eliminate rules that occur too rarely to be profitable, ensuring businesses focus only on frequent items.
- **Confidence Metric:** Confidence measures how frequently items in the second set (Y) appear in transactions that already contain the first set (X).
- **Reliability Estimate:** It provides an estimate of the conditional probability of Y given X, showing how reliable the inference made by the rule actually is.
- **Strong Rules:** A rule is only considered "strong" if it satisfies both a user-defined minimum support and a minimum confidence threshold simultaneously.

15. Outline the Pros and Cons of using the Apriori Algorithm.

The Apriori Algorithm is a classic breadth-first search method for finding frequent itemsets with specific operational trade-offs.

- **Simplicity and Implementation:** One of the main pros is that the algorithm is very simple and easy to implement compared to more complex tree-based methods.
- **Large Dataset Handling:** It works effectively with large datasets by identifying frequent items and pruning the search space systematically.
- **Business Value:** It is highly effective for generating strong association rules that provide actionable insights for retail and recommendation systems.
- **Database Scans:** A major con is that it requires many full scans of the database to count support, which is computationally expensive.
- **High-Dimensional Issues:** Performance decreases significantly with high-dimensional data as the number of candidate itemsets can become unmanageably large.

2-Mark Questions

16. What is the Knowledge Discovery in Databases (KDD) process?

- **Definition:** KDD is the complete process of extracting useful information and understandable patterns from large amounts of raw data.
- **Iterative Nature:** It involves a sequence of steps including problem definition, data preparation, modeling, and evaluation to turn data into knowledge.

17. Define Predictive Modeling.

- **Forecast Goal:** It is the process of using known historical results to create and validate a model that forecasts what might happen in the future.
- **Core Examples:** Common examples include classification and regression, used to predict things like student placement or the sale price of a property.

18. What is the Entity Identification problem in data integration?

- **Matching Conflict:** This problem occurs when an analyst must determine if different identifiers, like "student_id" and "student_number," refer to the same entity.
- **Metadata Solution:** The solution involves using metadata (data about data) as a reference to avoid errors and ensure correct mapping during integration.

19. Define Database Segmentation.

- **Clustering Basis:** It is a technique based on the concept of clustering where data is grouped into similar records that share common properties.
- **Unsupervised Learning:** It falls under unsupervised learning because the data is not pre-labeled before the segmentation process begins.

20. State the Apriori Principle.

- **Subset Rule:** The principle states that if an itemset is frequent, then all of its subsets must also be frequent based on the support threshold.
- **Infrequent Pruning:** Conversely, if an itemset is infrequent, all of its supersets will also be infrequent and do not need to be tested.

21. What constitutes a Strong Rule in association mining?

- **Threshold Satisfaction:** A rule is strong if it meets or exceeds both the minimum support and minimum confidence thresholds set by the user.
- **Reliability:** These rules are considered valid and useful for making business decisions because they represent high-probability relationships.

22. Define a Frequent Itemset.

- **Support Threshold:** A frequent itemset is a collection of items whose support count is greater than or equal to a pre-determined minimum support threshold.
- **Step One:** Finding these itemsets is the first and most computationally expensive step in association rule mining.

23. What is Deviation Detection?

- **Outlier Focus:** This technique focuses on identifying outliers that represent a significant departure from previously known expectations or norms.
- **Core Uses:** It is primarily performed using statistics and visualization for tasks like fraud detection and quality control in manufacturing.

10 Marks (Pakka)

10 → Frequent Item Set Generation / Rule Paper & Line

(or) FP-Growth Algorithm

Date

① 3-item sets

$\{B, S, O\}$

↓

1) $\{B, O\} \Rightarrow \{B\} \rightarrow \{O\} = 2/4 = 50\%$

$\{O\} \rightarrow \{B\} = 2/2 = 100\% \checkmark$

2) $\{B, S\} = \{B\} \rightarrow \{S\} = 3/4 = 75\%$

$\{S\} \rightarrow \{B\} = 3/4 = 75\% \checkmark$

3) $\{S, O\} \Rightarrow \{S\} \rightarrow \{O\} = 2/2 = 100\% \checkmark$

$\{O\} \rightarrow \{S\} = 2/4 = 50\%$

4) $\{B, S, O\} \Rightarrow \{B\} \rightarrow \{S, O\} = 2/4 = 50\%$

$\{S, O\} \rightarrow \{B\} = 2/2 \Rightarrow 100\% \checkmark$

5) $\{S, O\} \rightarrow \{B, S\} = 2/2 \Rightarrow 100\% \checkmark$

$\{B, S\} \rightarrow \{O\} = 2/3 \Rightarrow 67\%$

6) $\{S\} \rightarrow \{B, O\} = 2/4 \Rightarrow 50\%$

$\{B, O\} \rightarrow \{S\} = 2/2 \Rightarrow 100\% \checkmark$

Calculate Confidence

②

G

Item	Support	L ₂	Support	C ₃	Sup
B, H	1 X	B, O	2	B, S	2
B, O	2	B, S	3	B, S, H	1 X
B, S	3	H, S	2	H, O, S	0 X
H, O	0 X	O, S	2		
H, S	2				
O, S	2				

②

min-Support = 33.33%

min-Confidence = 60%

TID	Items
T1	H, B, K
T2	H, B
T3	H, C, D
T4	D, K
T5	O, K
T6	H, C, D

$33.33 \times 6 = 200$

(Ans) 4 rules $L_3 = \{H, C, D\} \Rightarrow 2$

Supporting Algorithm

(2)

Define what is Confidence & min Support

Step 2

Step 1: C1 & Y	Item	Sup-Count	Step 2	Item	Sup-Count	
	H	4		H, B	2	
	B	2		H, K	1	
	K	2		H, C	2	
	C	3		H, D	2	
	D	4		B, K	1	
				B, C	0	X
				B, D	0	X
				K, C	0	X
				K, D	1	X
				C, D	3	

Step-3	G3	Item	Sup-Count	
		H, B, C	0	
		H, B, D	0	
		H, C, D	2	

L3	Item	Sup-Count
	H, C, D	2

→ Association rules

 $\{H, C, D\}$

↓

$\{H\} = \{C, D\} \Rightarrow 2/4 = 50\% \quad X$

$\{C\} = \{H, D\} \Rightarrow 2/3 = 66.66\% \quad \checkmark$

$\{D\} \rightarrow \{C, H\} \Rightarrow 2/4 = 50\% \quad X$

$\{H, C\} \rightarrow \{D\} = 2/2 \Rightarrow 100\% \quad \checkmark$

$\{H, D\} = \{C\} = 2/2 \Rightarrow 100\% \quad \checkmark$

$\{C, D\} \rightarrow \{H\} = 2/3 \Rightarrow 66.66\% \quad \checkmark$

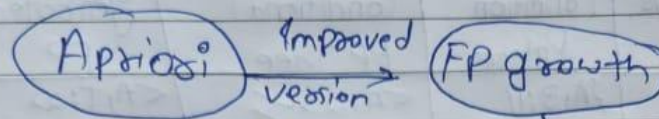
APRIORI Algo

Till FP growth
Frequent Pattern
FP growth

Paper & Line

Date

9 Feb 2025



No Need for
candidate list

→ FP Tree

1. Asparagus (A), Corn (C), Beans (B), Tomatoes (T)
& Squash (S)

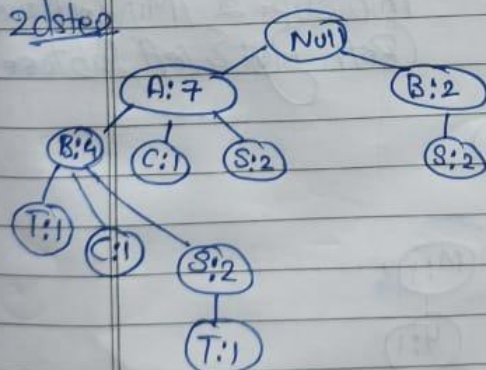
1st step

Transaction ID	List of items in the transactions	item / Support Count
T1	B, A, T	
T2	A, C	
T3	A, S	
T4	B, A, C	B → 6
T5	B, S	A → 7
T6	A, S	S → 6
T7	B, S	C → 2
T8	B, A, S, T	T → 2
T9	B, A, S	

Any Count less than 2 Remove it

→ A(7) → B(6) S(6) → C(2) T(2)

2nd step



T1: ABT

T2: AC

T3: AS

T4: ABC

T5: BS

T6: AS

T7: BS

T8: ABST

T9: ABS

④

Min Count 2

3rd Step	Items	Condition Patterns	Conditional FP Tree	Generated FP
T	{A, B:1}	{A, B:1}	$\langle A:2 \rangle \checkmark$ $\langle B:2 \rangle \checkmark$ $\langle A, B:2 \rangle \times$	$\langle A, T:2 \rangle$ $\langle B, T:2 \rangle$ $\langle A, B, T:2 \rangle$
C	{A, B:1}	{A, B:1}	$\langle A:2 \rangle \checkmark$ $\langle A:1 \rangle \times$ $\langle A, B:1 \rangle \times$	$\langle A, C:2 \rangle$
S	{A:2}	{A:2}	$\langle A:1 \rangle \checkmark$	$\langle A, S:1 \rangle$
B	{A:1}	{A:1}	$\langle B:2 \rangle \checkmark$ $\langle A, B:2 \rangle \checkmark$	$\langle B, S:1 \rangle$ $\langle A, B, S:2 \rangle$
			$\langle A:1 \rangle \checkmark$	$\langle A, B:1 \rangle$

If Sub trees

Don't merge

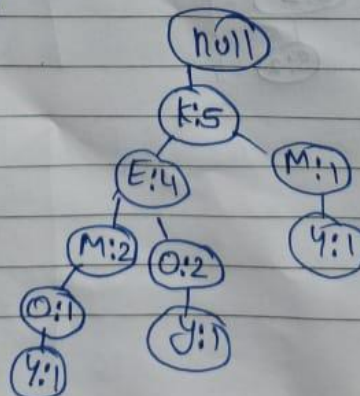
TID	Items
T1	EKMNOY
T2	DEKNOY
T3	AEKM
T4	CKMUY
T5	CEIKOO

min Sup=3
(only left subtree)

Items	TID
I1 I2 I5	T1
I2 I4	T2
I2 I3	T3
I1 I2 I4	T4
I1 I3	T5
I2 I3	T6
I1 I3	T7
I1 I2 I3 I5	T8
I1 I2 I3	T9

min Sup=2 / Min (confidence)
(Both right & left subtree)

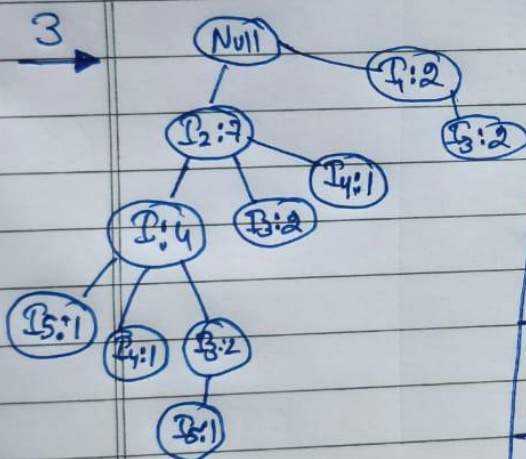
Items	Sup-count
K	5
E	4
M	3
O	3
Y	3



Paper & Line

Date

Items	Conditional Pattern base	Conditional FP Tree	FP generated
Y	{KEMO:1} {KEO:1} {KM:1}	$\langle K:3 \rangle$	$\langle K, Y:3 \rangle$
O	{KEM:1} {KE:2}	$\langle K, E:3 \rangle$	$\langle K, O:3 \rangle$ $\langle E, O:3 \rangle$ $\langle K, E, O:3 \rangle$
M	{KE:2} {K:1}	$\langle K:3 \rangle$	$\langle K, M:3 \rangle$
E	{K:4}	$\langle K:4 \rangle$	$\langle K, E:4 \rangle$



item	Conditional Pattern base	Conditional FP Tree	FP generated
I5	{I2, I4:1} {I2, I4, I3:1}	{I, I2:2}	$\langle I, I5:2 \rangle$ $\langle I, I5:2 \rangle$ $\langle I, I2, I5:2 \rangle$
I4	{I2, I4:1} {I3:1}	{I2:2}	$\langle I2, I4:2 \rangle$
I3	{I2, I4:2} {I2:2}/ {I4:2}	{I2:4, I4:2} / {I2:2}	$\langle I2, I3:4 \rangle$ $\langle I, I3:4 \rangle$ $\langle I, I2, I3:2 \rangle$
I1	{I2:4}	{I2:4}	$\langle I2, I1:4 \rangle$

I, I